

CSC 307

Operating System

Examination: Semester Examination	Semester: Second Semester
Year: 2024	Duration: 2 hours
Total Questions: 5	

Instructions: Answer ALL questions. All questions carry equal marks unless otherwise stated.

Question 1

- (a) Define an Operating System and explain its four main functions. (8 marks)
- (b) Differentiate between batch processing, multiprogramming, and time-sharing systems. (7 marks)

Question 2

- (a) Explain the process lifecycle with a state diagram. (10 marks)
- (b) What is a Process Control Block (PCB)? List and explain five fields in a PCB. (5 marks)

Question 3

Consider the following set of processes with their arrival times and burst times:

Process Arrival Time Burst Time

P1 0 8

P2 1 4

P3 2 9

P4 3 5

Calculate the average waiting time and average turnaround time using:

- (a) First-Come-First-Served (FCFS) (8 marks)
- (b) Shortest Job First (SJF) (7 marks)

